English-Dutch Machine Translation

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1 Data Exploration

1.1 Corpus Overview. The English-Dutch Europarl corpus contained 1,997,775 aligned sentence pairs. The English and Dutch files contained the same number of lines, confirming that sentences could be paired according to their line positions.

Initial inspection identified 15,638 empty English lines and 3,392 empty Dutch lines. Pairs containing an empty sentence on either side were excluded from the exploratory statistics because they did not represent complete translations. This left 1,978,745 valid sentence pairs. No lines beginning with an XML tag were found.

The corpus also contained 40,320 exact duplicate pairs, representing approximately 2% of the data. These were retained in the experiments, although their presence is a limitation because identical pairs may occur in different data splits and could slightly increase the evaluation scores.

1.2 Sentence Length Analysis. Table 1 summarizes the English and Dutch sentence lengths after pairs containing empty sentences were excluded, and Figures 1 to 3 illustrate these distributions.

Statistic	English	Dutch
Mean words	25.02	25.48
Median words	22	23
Standard deviation	14.90	15.10
Mean characters	148.92	163.49
Median characters	133	146
Maximum words	668	583

Table 1: Statistics of sentences

The English and Dutch sentences had similar word counts. Dutch translations contained only 0.46 more words on average. Dutch sentences were longer in 47.69% of the aligned pairs, while 12.61% contained the same number of words.

The correlation between English and Dutch word counts was 0.928, indicating a strong relationship between source and target length. Longer English sentences generally corresponded to longer Dutch translations. Nevertheless, individual translations did not always contain the same number of words because a single expression in one language may require several words in the other.

The character-level difference was larger. Dutch sentences contained approximately 163 characters on average, compared with 149 characters for English. This suggests that Dutch tends to use longer word forms, potentially because of its inflection and compound-word formation.

The maximum sentence lengths of 668 English words and 583 Dutch words showed that the corpus contained a small number of extreme outliers. The length-distribution graph was therefore clipped at 80 words so that the distribution of typical sentences remained visible. These outliers also motivated the later use of maximum sequence lengths in the neural models.

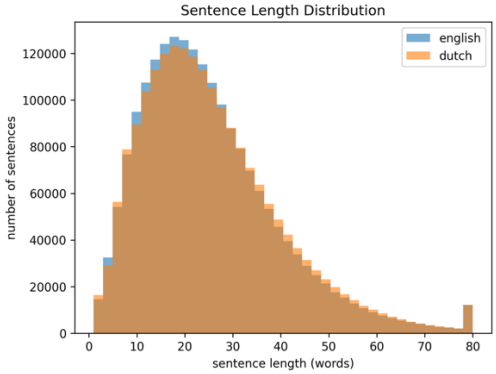


Figure 1: Distribution of English and Dutch sentence lengths

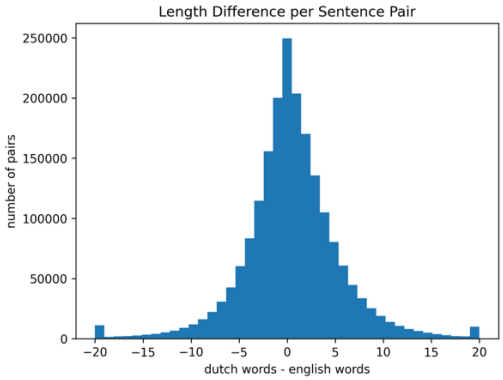


Figure 2: Difference between Dutch and English word counts

1.3 Vocabulary and Word Frequency. Using lowercase whitespace tokenization, the English corpus contained approximately 49.50 million tokens and 275,650 unique tokens. The Dutch corpus contained approximately 50.42 million tokens and 495,226 unique tokens. Therefore, the observed Dutch vocabulary was approximately 1.8 times larger than the English vocabulary.

Rare words formed a substantial proportion of both vocabularies. A total of 133,995 English tokens and 271,086 Dutch tokens occurred only once. These values represent

approximately 48.6% of the English vocabulary and 54.7% of the Dutch vocabulary. This indicated that including every observed token would create a very large model and that Dutch was more likely to be affected by unknown words.

Despite the large vocabularies, the most frequent words covered most of the corpus:

Coverage	English	Dutch
Top 100 words	52.1%	53.6%
Top 1000 words	75.7%	75.5%
Top 10000 words	94.8%	92.5%

Table 2: Vocabulary coverage

The high coverage of frequent words supported limiting the later word-based models to a vocabulary of 20,000 tokens. The slightly lower Dutch coverage was consistent with its greater vocabulary diversity.

The most frequent tokens were function words such as "the," "of," and "to" in English and "de," "van," and "het" in Dutch. Although these words carry limited topical information, they are essential for grammatical translation. This finding supported retaining stop words during preprocessing.

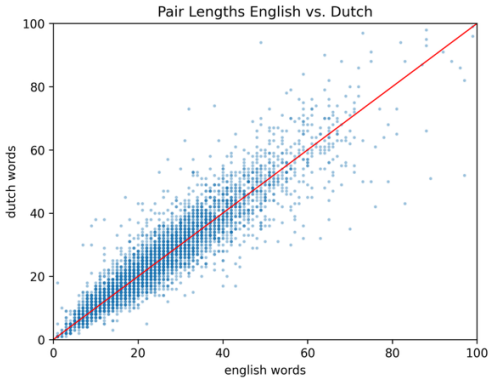


Figure 3: Sentence lengths (whitespace tokens) of aligned English–Dutch pairs, on a random sample of 5000 pairs from the cleaned corpus. The red line marks equal length ($y = x$); axes are clipped at 100 tokens. Lengths are strongly correlated across the full corpus ($r = 0.928$), with Dutch slightly longer on average (+0.46 tokens) and the spread widening for longer sentences.

1.4 Data Sampling. Training neural translation models on almost two million sentence pairs would require substantial time and memory. Therefore, 10% of the parallel corpus was randomly selected without replacement. The English and Dutch sentences were sampled jointly to preserve their alignment, and a fixed seed of 42 was used for reproducibility.

The resulting raw sample contained 199,778 aligned pairs. Its average English sentence length was approximately 24.8 words, close to the full-corpus average of 25.02. This indicated that the sample retained the general sentence-length characteristics of the original corpus. After the preprocessing described in Section 2, 197,853 pairs remained for model development.

2 Preprocessing

2.1 Data Sampling and Pair Alignment. The original Europarl corpus contained 1,997,775 aligned English-Dutch sentence pairs. Since the complete corpus was too large for the available computational resources, 10% of the aligned pairs were randomly selected before model development. The same sampled indices were applied to both language files so that every English sentence remained paired with its corresponding Dutch translation. Sampling was performed without replacement using a fixed random seed of 42. This made the selection reproducible and reduced the possibility of obtaining a biased subset by taking only one consecutive part of the corpus. The initial sample contained 199,778 sentence pairs.

Preprocessing was performed on each English-Dutch pair jointly. If either side of a pair was removed, its corresponding sentence was also removed. This was necessary because deleting a line from only one language would shift the remaining lines and destroy the parallel alignment required for supervised machine translation.

2.2 Applied Preprocessing. First, leading and trailing whitespace was removed, and repeated whitespace inside each sentence was replaced with a single space. Consistent whitespace was useful because the later word-based model used whitespace tokenization. It also prevented irregular spacing from producing inconsistent textual representations.

Sentence pairs were removed when either the English or Dutch sentence was empty. An empty source sentence provides no information for the encoder, while an empty target sentence provides no translation for the decoder to learn. Removing both sides of such pairs preserved corpus alignment.

Lines beginning with the character < were also filtered as a precaution, although no such lines were found in the sampled corpus. Generally, including them would introduce tokens unrelated to translation and could cause the model to learn structural corpus markers instead of linguistic correspondences.

Finally, both English and Dutch sentences were converted to lowercase. Lowercasing reduced vocabulary size by treating forms such as "Parliament" and "parliament" as the same word. This was particularly useful because the later models used a limited vocabulary: unnecessary capitalization variants could otherwise increase the number of rare or unknown tokens. After invalid pairs were removed, 197,853 aligned sentence pairs remained. These cleaned files were then used for the training, validation, and test split.

2.3 Preprocessing Not Applied. Stop-word removal was not applied because frequent function words remain important in translation. Words such as articles, pronouns, prepositions, and auxiliary verbs contribute to grammatical structure and word order. Removing them could produce incomplete sentences and prevent the model from learning how these words should be translated. Negation words such as "not," "no," and "niet" are especially important because removing them can reverse the meaning of a sentence.

Stemming and lemmatization were also not used. Although these methods can reduce vocabulary size, a translation model must learn and generate the correct surface forms of words. Dutch in particular contains inflection and productive compound formation. Replacing words with stems or lemmas would remove grammatical information and would make the training targets different from natural Dutch sentences.

Punctuation was retained because it provides information about

sentence boundaries, abbreviations, quotations, and grammatical structure. It can also affect meaning and translation style. Tokenization and sequence padding were therefore left to the TextVectorization layers used during model construction rather than being performed permanently in the preprocessing files.

Unicode compatibility normalization was considered, but an inspection of the selected corpus found no sentences that changed under NFKC normalization. Consequently, it was not applied because it would add an unnecessary transformation. Dutch characters and diacritics were retained in their original form. Overall, the preprocessing was intentionally conservative: noise and corpus metadata were removed, while information needed to learn complete and grammatically meaningful translations was preserved.

3 Neural Machine Translation

3.1 Experimental Setup. The 197,853 preprocessed sentence pairs were shuffled using a fixed random seed of 42 and divided into training, validation, and test sets. The training set contained 138,498 pairs (70%), the validation set contained 19,785 pairs (10%), and the test set contained 39,570 pairs (20%). The validation set was used for hyperparameter selection and early stopping, while the test set was reserved for the final evaluation.

Separate TextVectorization layers converted English and Dutch words into integer IDs. Each vocabulary was limited to the 20,000 most frequent tokens to reduce memory usage and the size of the model's output layer. Less frequent words were represented by an unknown token. English sequences were limited to 55 tokens and Dutch sequences to 57 tokens, with shorter sequences padded to a consistent length.

The target sentences were surrounded by startmark and endmark tokens. The start marker indicated when the decoder should begin generating a translation, while the end marker taught it when to stop. During training, the decoder input contained the target sequence without its final token, and the expected output was shifted one position to the left. This teacher-forcing arrangement trained the decoder to predict the next target word from the correct preceding words.

3.2 Model Architecture. A word-based recurrent encoder-decoder architecture was developed. The encoder transformed each source token into a 128-dimensional embedding and processed the sequence using a Gated Recurrent Unit (GRU). The encoder's final hidden state was used to initialize a second GRU in the decoder. At every decoder position, a dense output layer produced one score for every word in the target vocabulary.

GRUs were selected because they can preserve information across sequential input and are computationally simpler than LSTMs due to their smaller number of gates. This was suitable for the available computational resources while still allowing the model to learn word order and dependencies between source and target sentences. However, because the encoder compressed the complete source sentence into one fixed-size state, the architecture was expected to have difficulty retaining all information from long sentences.

The model was trained using sparse categorical cross-entropy and the Adam optimizer. Gradient clipping with a norm of 1.0 was used to reduce the risk of exploding gradients. Greedy decoding was used during inference, meaning that the highest-scoring word was selected at each decoding step.

3.3 Evaluation Metrics. BLEU and chrF were selected as the two evaluation metrics. BLEU measures exact overlap between the word sequences in a model translation and a human reference. It therefore indicates whether the model generates matching words and phrases. However, it can give limited credit to translations that use different word forms.

chrF compares character sequences and can recognize partial similarity between related or inflected words. This is useful for Dutch because of its inflection and productive compound words. Using both metrics provided complementary views of translation performance. Validation scores were used for model selection, while the final reported performance was calculated on the previously unseen test set.

3.4 Hyperparameter Tuning. Manual tuning was performed using a fixed subset of 20,000 training pairs and 3,000 validation pairs to reduce computational cost. BLEU and chrF were calculated using the same 500 validation examples in every tuning experiment. Only one main setting was changed between experiments whenever possible.

Embedding	GRU size	Learning rate	Epochs	Val. loss	Val. acc
Random	256	0.001	3	4.8065	25.40%
Random	256	0.001	5	4.5864	27.80%
Random	128	0.001	5	4.7081	26.09%
Random	256	0.002	max. 5	4.5739	28.42%
Word2Vec	256	0.002	max. 5	4.5848	28.20%
FastText	256	0.002	max. 5	**4.5630**	**28.53%**

Table 3: Val Loss & Acc based on different configurations

BLEU	chrF
1.92	11.96
2.73	14.91
1.52	13.18
2.65	**17.09**
2.77	15.11
2.91	13.72

Table 4: Corresponding BLEU & chrF metrics

As it is shown in both Table 3 and Table 4, increasing training from three to five epochs improved both validation metrics, indicating that three epochs were insufficient. Reducing the GRU size from 256 to 128 lowered BLEU and chrF, suggesting that the smaller hidden state had insufficient capacity to represent the sentence information. The 256-unit GRU was therefore retained.

Increasing the learning rate from 0.001 to 0.002 produced the highest chrF score among the randomly initialized models. Early stopping with a patience of one epoch was added because validation loss began to stop improving while training loss continued to decrease. The best weights were restored after training.

3.5 Embedding Comparison. Three embedding initializations were evaluated: randomly initialized trainable embeddings, Word2Vec, and FastText. Word2Vec and FastText embeddings were trained on the tuning corpus and used to initialize the encoder and decoder embedding layers. These embeddings remained trainable during translation training.

Based on the results from Table 3 and Table 4, FastText achieved the best validation loss, word accuracy, and BLEU, but its chrF score was lower than that of the randomly initialized embeddings. Word2Vec provided a small BLEU improvement over the selected random model but also produced a lower chrF score. Therefore, neither alternative embedding consistently improved both evaluation metrics.

The final model used randomly initialized trainable embeddings because they achieved the highest validation chrF score and required no separate embedding-training stage. The result also suggests that

embeddings learned directly for the translation objective were competitive with embeddings first trained using local word-context prediction.

3.6 Final Results. The selected configuration used 128-dimensional trainable embeddings, 256 GRU units, a vocabulary limit of 20,000 words, Adam with a learning rate of 0.002, a batch size of 32, and a maximum of five epochs. Early stopping restored the model from the epoch with the lowest validation loss.

Translation	Best Epoch	Val. loss	Val. acc	BLEU	chrF
English to Dutch	3	3.4846	36.64%	5.78	23.81
Dutch to English	3	3.5647	37.74%	6.60	28.00

Table 5: Final results from test set

For English-to-Dutch translation, validation loss improved until epoch 3 and increased slightly in epoch 4, indicating the beginning of overfitting. The restored epoch-3 model achieved a test BLEU score of 5.78 and a chrF score of 23.81.

The same architecture and hyperparameters were then used after reversing the source and target languages. Dutch-to-English translation achieved a BLEU score of 6.60 and a chrF score of 28.00. This represents improvements of 0.82 BLEU and 4.19 chrF compared with English-to-Dutch translation. One possible explanation is that the limited English target vocabulary provided better coverage than the Dutch target vocabulary. Dutch contains more inflected and compound forms, which can become unknown tokens when the vocabulary is limited.

3.7 Sentence Lengths and Characteristics Analysis. To measure the effect of source-sentence length, the English-to-Dutch test set was divided into three groups: short sentences containing at most 10 words, medium sentences containing 11–20 words, and long sentences containing more than 20 words.

Sentence length	Sentences	BLEU	chrF
Short (<= 10 words)	5339	**13.08**	**31.07**
Medium (11 - 20 words)	11983	6.99	25.83
Long (> 20 words)	22248	5.10	22.93

Table 6: Metrics regarding different sentence lengths

Both evaluation metrics decreased consistently as sentence length increased. Short sentences achieved a BLEU score of 13.08 and a chrF score of 31.07, while long sentences achieved only 5.10 BLEU and 22.93 chrF. From the short to the long group, BLEU decreased by 7.98 points and chrF decreased by 8.14 points. The results therefore provide clear evidence that sentence length negatively affected translation quality.

Short sentences generally contained fewer relationships for the model to preserve and often consisted of frequent parliamentary expressions, greetings, or simple statements. In contrast, long sentences frequently contained several clauses, rare terms, and more complex grammatical relationships. Because the encoder represented the entire source sentence using a single fixed-size hidden state, information could be lost as sentence length increased. This resulted in omitted details, repeated phrases, unknown tokens, and generic translations. An attention mechanism may reduce this limitation by allowing the decoder to refer directly to different encoder positions while generating each target word.

3.8 Character-based Model. A character-level English-to-Dutch model was trained using the same GRU encoder-decoder architecture as the word model. It used 128-dimensional embeddings, 256 GRU units, maximum sequence lengths of 320 English and 322 Dutch characters, and the complete training set. The best model occurred at epoch 5 with a validation loss of 0.9987.

The character model achieved **2.36 BLEU** and **20.39 chrF**, compared with **5.78 BLEU** and **23.81 chrF** for the word model. Character generation avoided frequent [UNK] tokens and produced complete words, but the substantially longer sequences made it difficult for the fixed encoder state to preserve meaning. Its translations frequently contained generic phrases, repetition, and omitted details. Therefore, the word-based model performed better overall.

4 Neural Machine Translation with Attention

Additive attention was added to the English-to-Dutch GRU encoder-decoder. Unlike the original model, the encoder returned an output for every source position. At each decoding step, attention calculated a weighted combination of these outputs and concatenated it with the decoder state before predicting the next Dutch word. The dataset, embeddings, GRU size, optimizer, batch size, and maximum epochs remained unchanged. The best attention model occurred at epoch 3 with a validation loss of 2.9660 and validation accuracy of 44.15%.

Model	Val.loss	Val.acc	Test BLEU	Test chrF
Without attention	3.4846	36.64%	5.78	23.81
With attention	2.9660	44.15%	11.59	33.55

Table 7: Comparison of the models with attention and without

Attention improved BLEU by 5.81 points and chrF by 9.74 points. It reduced the fixed-state bottleneck by allowing the decoder to refer directly to relevant source positions. The resulting translations preserved more content and produced fewer generic repetitions, although unknown words, omissions, and semantic errors remained, especially in longer sentences.

Figure 4 visualizes the attention weights for the source sentence “I would also like to thank you, minister.” Brighter cells indicate greater attention. The decoder concentrated strongly on “you” and “minister” while generating the corresponding Dutch phrase. However, it translated “minister” as “mevrouw de voorzitter,” showing that attention can identify a relevant source position without guaranteeing the correct target word.

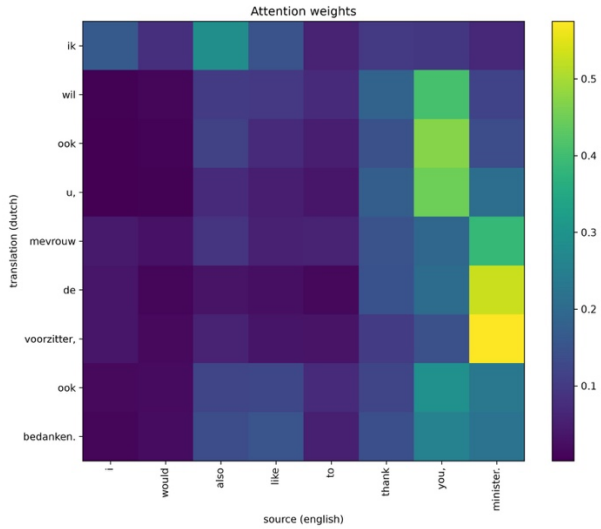


Figure 4: Attention weights for an English to Dutch sample translation

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